THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 11: **The World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** a precise step-by-step procedure that solves a problem	definition of an algorithm
2	**(a)** units must be added to units, tens to tens, and so on	place-value grouping
3	**(a)** gcd(n, m − n)	every common divisor of m and n also divides m − n
4	**(a)** Both A and R are true and R is the correct explanation of A	the set of common divisors does not change at each reduction
5	**(b)** 2	carries by column: [1, 1, 0]
6	**(b)** 4	carries by column: [1, 1, 1, 1]
7	**(b)** 18	repeated reduction ends at gcd(18, 0)
8	**(a)** 5	the chain of reductions ends at gcd(5, 0)
9	**(d)** 6	divisors(18) = [1, 2, 3, 6, 9, 18]
10	**(b)** 2, 3, 5	prime divisors have exactly two divisors; 1 is not prime

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	An algorithm is a step-by-step procedure that leads to the answer of a problem (1). Example: the digit-by-digit addition algorithm / Euclid's gcd algorithm (1).	1+1
12	Sum $= 10363$ (1). Carries: units (7+6=13), tens (8+7+1=16), hundreds (9+3+1=13) and thousands (5+4+1=10) — all four columns; the last carry is written as the leading 1 (1).	1+1
13	After the leftmost column a carry of 1 may remain; if not written, a place value is lost (1). E.g. $95 + 17 = 112$, but without Step 5 we get 12 (1).	1+1
14	Check j = 1, 2, …, 20: 1, 2, 4, 5, 10, 20 divide 20 (1). List $= [1, 2, 4, 5, 10, 20]$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Units $8 + 9 = 17$: write 7, carry 1 (1). Tens $6 + 7 + 1 = 14$: write 4, carry 1 (1). Hundreds $9 + 4 + 1 = 14$: write 4, carry 1; thousands $7 + 5 + 1 = 13$: write 3, carry 1; final carry written: sum $= 13447$ (1).	1+1+1
16	The numbers are aligned on the right (1). Once the digits of the shorter number are exhausted, the missing places are treated as 0 and only carries are added (1), e.g. $99\,999 + 123 = 100\,122$ is handled correctly (1).	1+1+1
17	Each number is grouped into units, tens, hundreds … (1). We add each group separately (1). Whenever a group of ten forms in a column, it is carried to the next higher place (1).	1+1+1
	OR: Counting: $3347 + 2729 = 6076$ dots, about 6076 s, over 1 h 40 min (1). Column method: 4 columns × 15 s $= 60$ s (1). Column method is far faster; its time grows with the number of digits (1).	1+1+1
18	After j=1: [1]; j=2: [1,2]; j=4: [1,2,4]; j=7: [1,2,4,7] (2). Final: [1, 2, 4, 7, 14, 28] (1).	2+1
	OR: divisors(36) $= [1, 2, 3, 4, 6, 9, 12, 18, 36]$; divisors(60) $= [1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60]$ (1). Common $= [1, 2, 3, 4, 6, 12]$ (1). gcd $= 12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Steps: align right; add the rightmost digits, write the units digit and set carry (0 or 1); add next column with carry; repeat; write a final carry (2). Execution: $5 + 8 = 13 \to 3$, c=1; $8 + 4 + 1 = 13 \to 3$, c=1; $7 + 7 + 1 = 15 \to 5$, c=1; $6 + 2 + 1 = 9 \to 9$, c=0; sum $= 9533$ (2). Step 5 writes any leftover carry; without it, e.g. $9999 + 1$ would give 0000 (1).	2+2+1
	OR: Algorithm: start with empty list; for each $j = 1, 2, \ldots, n$, if j divides n, add j at the end (2). For 24: $[1, 2, 3, 4, 6, 8, 12, 24]$ (2). j is tested in increasing order and appended, so the list is increasing (1).	2+2+1

20	Steps: get divisors(m), divisors(n); empty common list; for each x in divisors(m), if x in divisors(n) add to common; report the rightmost element (2). divisors(30) $= [1, 2, 3, 5, 6, 10, 15, 30]$; divisors(45) $= [1, 3, 5, 9, 15, 45]$ (1). Common $= [1, 3, 5, 15]$ (1); gcd $= 15$ (1).	2+1+1+1
	OR: (i) Combine the scans: check j up to $\max(m, n) = 60$ instead of $m + n = 96$ (2). (ii) Directly compute common divisors, using up to $\min(m, n) = 36$ (2). (iii) Drop lists: keep only most-recent-common-divisor, scanning $k = 2$ to 36 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 11: **The World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** adding two single-digit numbers	algorithms are built from basic steps we can already perform
2	**(b)** counting dots needs about 100 times as many dots, but the column method needs only 2 more columns	dots grow with the value, columns grow with the number of digits
3	**(d)** n = 0	gcd(m, 0) = m
4	**(a)** Both A and R are true and R is the correct explanation of A	99 mod 2 = 1 in one step
5	**(d)** 3	carries by column: [1, 1, 1]
6	**(d)** 75	repeated reduction ends at gcd(75, 0)
7	**(a)** 7	the chain of reductions ends at gcd(75, 0)
8	**(b)** 3	each reduction replaces (m, n) by (n, m mod n); ends at gcd(4, 0)
9	**(c)** 9	divisors(36) = [1, 2, 3, 4, 6, 9, 12, 18, 36]
10	**(a)** 2, 3, 5	9 = 3 × 3 is not prime

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	All $j = 1, 2, \ldots, n$ — that is n numbers (1). We test j in increasing order and append at the end, so the list stays in increasing order (1).	1+1
12	divisors(48) $= [1, 2, 3, 4, 6, 8, 12, 16, 24, 48]$, divisors(18) $= [1, 2, 3, 6, 9, 18]$ (1). Common $= [1, 2, 3, 6]$; gcd $= 6$ (1).	1+1
13	Both lists of divisors are found in one scan of j from 1 to $\max(m, n)$ (1), so $\max(m, n)$ numbers are checked instead of $m + n$ (1).	1+1
14	A common divisor divides both numbers, so it cannot be larger than the smaller number (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Start with most-recent-common-divisor $= 1$; for each k from 2 to $\min(m, n)$, if k divides both, update it to k (1). $\min = 8$: $k = 2$ ✓ → 2; $k = 3$ ×; $k = 4$ ✓ → 4; $k = 5, 6, 7$ ×; $k = 8$ × (does not divide 12) (1). gcd $= 4$ (1).	1+1+1
16	It scans every value from 1 to $\min(m, n)$, and a 4-digit number is about 10 times as large as a 3-digit one (1, 1). Work is proportional to the value, not to the number of digits, so it is slow for large numbers (1).	1+1+1
17	Write $m = ad$, $n = bd$ for natural numbers a, b (1). $m - n = ad - bd = (a - b)d$ (1). Hence d divides $m - n$ (1).	1+1+1
	OR: Write $n = xd$ and $m - n = yd$ (1). $m = n + (m - n) = xd + yd = (x + y)d$ (1). So d divides m (1).	1+1+1
18	$(42, 30) \to (30, 12) \to (12, 18) \to$ swap $(18, 12) \to (12, 6) \to (6, 6) \to (6, 0)$ (2). Step 3 is applied 5 times (the swaps are Step 1); gcd $= 6$ (1).	2+1
	OR: Subtraction: $(9, 2) \to (2, 7) \to (7, 2) \to (2, 5) \to (5, 2) \to (2, 3) \to (3, 2) \to (2, 1) \to (1, 1) \to (1, 0)$: Step 3 is applied 6 times (2). Division: $9 \bmod 2 = 1$: $(2, 1) \to (1, 0)$: 2 reductions (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Statement (1). Let d divide m and n: $m = ad$, $n = bd$, so $m - n = (a - b)d$ and d divides $m - n$ (2). Conversely let d divide n and $m - n$: $n = xd$, $m - n = yd$, so $m = (x + y)d$ (1). The sets of common divisors are equal, hence the gcds are equal (1).	1+2+1+1
	OR: $(825, 375) \to (375, 450) \to (450, 375) \to (375, 75) \to (75, 300) \to (300, 75) \to (75, 225) \to (225, 75) \to (75, 150) \to (150, 75) \to (75, 75) \to (75, 0)$ (2). Step 3 used 7 times; gcd $= 75$ (1). It is slow because repeated subtraction of a small number from a large one takes about as many steps as the quotient, e.g. gcd(99, 2) (2).	2+1+2

20	Replace subtraction by remainder: reduce gcd(m, n) to gcd(n, m mod n) until n = 0 (2). $2587392 = 157656 \times 16 + 64896$; $157656 = 64896 \times 2 + 27864$; $64896 = 27864 \times 2 + 9168$ (1). $27864 = 9168 \times 3 + 360$; $9168 = 360 \times 25 + 168$; $360 = 168 \times 2 + 24$; $168 = 24 \times 7 + 0$ (1). gcd $= 24$ (1).	2+1+1+1
	OR: (i) $81000 = 51000 \times 1 + 30000$; $51000 = 30000 \times 1 + 21000$; $30000 = 21000 \times 1 + 9000$; $21000 = 9000 \times 2 + 3000$; $9000 = 3000 \times 3 + 0$ ⇒ gcd $= 3000$ (3). (ii) The remainders are $125695, 111961, 13734, 2089, 1200, 889, 311, 267, 44, 3, 2, 1, 0$: 13 reductions, gcd $= 1$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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